CH5　薄壁容器

5-1. 薄壁容器的應力分析

一、內壓薄壁圓筒的應力分析

(一) 薄壁圓筒在內壓 P 作用下，圓筒壁上任一點將產生兩個方向的應力，如圖所示，一個是由內壓作用在封頭上的軸向拉應力而引起的軸向應力稱之為徑向應力 σ_x；另一個是由於內壓作用使圓筒均勻向外膨脹，在圓周切線方向產生的應力，稱爲環向應力σ_y表示。

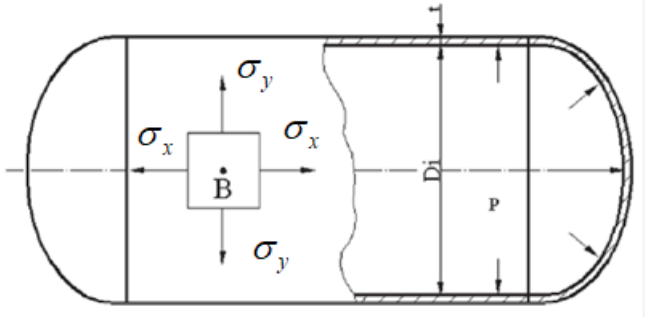
(二) 根據截面法，作一垂直於圓筒軸線的橫截面，將圓筒分成兩部分，根據平衡條件，內壓作用於焊接處的軸向外力必須與軸向應力作用於壁厚上的合力相等，即：

$$\pi R^2 P = \pi 2Rt\,\sigma_x$$

其中 R ：等於內半徑、P ：內壓力、t ：薄壁厚度

上式簡化後得：

$\sigma_x = \dfrac{PR}{2t}$ 同理可計算出 $\sigma_y = \dfrac{PR}{t}$



證明:

如圖所示考慮環向應力之自由體圖由力平衡方程式可得

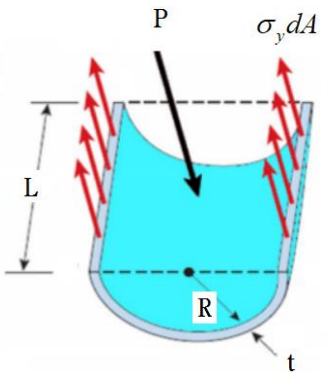
$$\sum F_y = 0 \Rightarrow \sigma_y dA - P(2RL) = 0 \Rightarrow \sigma_y (2Lt) - P(2RL) = 0$$

$$\sigma_y = \frac{PR}{t}$$

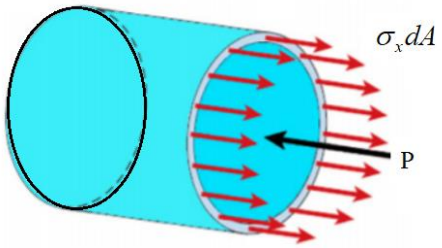
同理考慮軸向應力之自由體圖由力平衡方程式可得

$$\sum F_x = 0 \Rightarrow \sigma_x dA - P(\pi R^2) = 0 \Rightarrow \sigma_y (2\pi Rt) - P(R^2\pi) = 0$$

$$\sigma_x = \frac{PR}{2t}$$



環向應力
(Circumferential stress)



軸向應力
(longitudinal stress)

二、內壓薄壁球形容器的應力分析

以同樣的分析方法可以求得承受內壓作用下球形容器的應力，因球形容器是中心對稱，故殼體上各處的應力均相等，並且徑向應力σ_x與環向應力σ_y也相等，根據截面法可推出應力計算公式爲：

$$\sigma_x = \sigma_y = \frac{PR}{2t} \text{(球形容器)}$$

證明:

如圖所示考慮環向應力之自由體圖由力平衡方程式可得

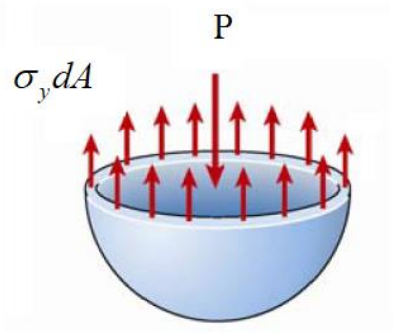
$$\sum F_y = 0 \Rightarrow \sigma_y dA - P(\pi R^2) = 0 \Rightarrow \sigma_y (2\pi Rt) - P(R^2\pi) = 0$$

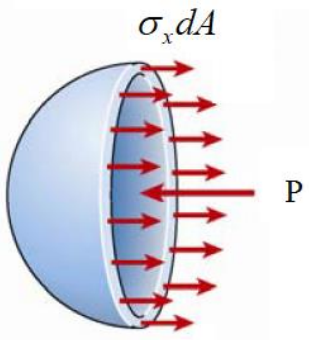
$$\sigma_y = \frac{PR}{2t}$$

同理考慮軸向應力之自由體圖由力平衡方程式可得

$$\sum F_x = 0 \Rightarrow \sigma_x dA - P(\pi R^2) = 0 \Rightarrow \sigma_y (2\pi Rt) - P(R^2\pi) = 0$$

$$\sigma_x = \frac{PR}{2t}$$





範例 1

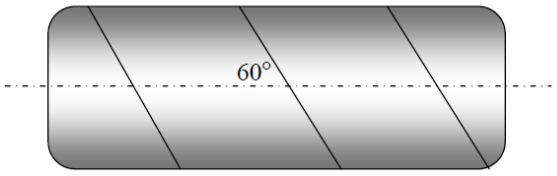
有一兩端封閉直徑 1.5 m 之圓筒型壓力容器，由厚度 15 mm 的鋼板捲成螺旋形銲接而成，螺旋形銲道與壓力容器軸向成60°。壓力容器承受均勻內壓力 1.2 MPa。

㈠求管壁上的軸向應力 σ_a 與圓周向應力 σ_h 以及面內（in-plane）最大剪應力 τ_{max}。（8 分）

㈡求垂直於銲道上的正向應力 σ_w 與平行於銲道上的剪應力 τ_w。（8 分）

㈢繪出此壓力容器承受內壓力 1.2 MPa 時容器壁應力狀態的莫耳圓（Mohr's circle），並標出㈠㈡小題求出之各個正向應力與剪應力在莫耳圓上所對應的位置、角度以及應力值。（10 分）

㈣若此壓力容器內壓力繼續增加直到容器壁破裂，由㈠㈡小題之解判斷破裂位置發生在銲道上或發生在非銲道位置？並述理由。（4 分）



【106 關務三等】

【解析】

(一)應力分析

環向應力 $\sigma_h = \frac{\text{Pr}}{t} = \frac{1.2\times 750}{15} = 60(MPa)$

軸向應力 $\sigma_L = \frac{\text{Pr}}{2t} = \frac{1.2\times 750}{2\times 15} = 30(MPa)$

(二)平面應力轉換公式

逆時針旋轉 $30^\circ \Rightarrow \theta = 30^\circ$ 代入

$$\sigma_w = \frac{\sigma_x+\sigma_y}{2} + \frac{\sigma_x-\sigma_y}{2}\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\Rightarrow \sigma_w = \frac{30+(60)}{2} + \frac{30-(60)}{2}\cos 2(30^0) + 0\times\sin 2(30^\circ) = 37.5(MPa)$$

$$\tau_w = -\frac{\sigma_x-\sigma_y}{2}\sin 2\theta + \tau_{xy}\cos 2\theta = 13(MPa)$$

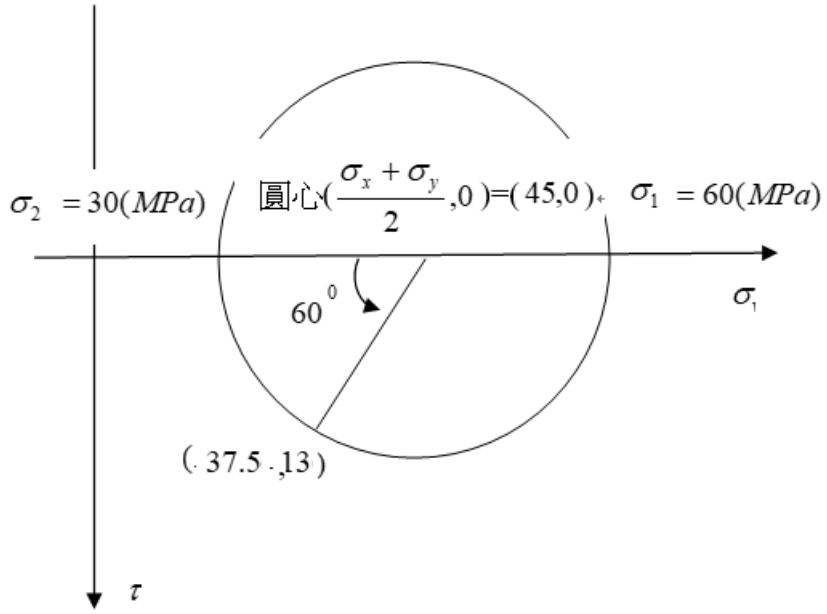
(三)求主應力及最大剪力

$\sigma_1 = \sigma_h = \frac{\text{Pr}}{t} = \frac{1.2\times 750}{15} = 60(MPa)$

$\sigma_2 = \sigma_L = \frac{\text{Pr}}{2t} = \frac{1.2\times 750}{2\times 15} = 30(MPa)$

圓心$(\frac{\sigma_x+\sigma_y}{2},0)=(45,0)$

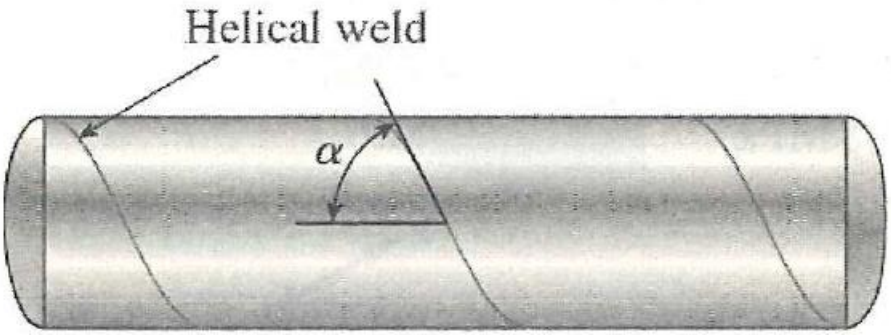
半徑$R == \frac{\sigma_1 + \sigma_2}{2} = 15(MPa)$



範例 2

A steel tank is constructed with a helical weld that makes an angle $\alpha = 60^\circ$ with the longitudinal axis. The vessel has inner radius $r = 2m$ and wall thickness $t = 25mm$. The steel has Young's modulus $E = 200\ \text{GPa}$ and Poisson's ratio $\upsilon = 0.3$. The internal pressure p is 600 kPa..

Determine the following quantities for the cylindrical part of the tank: (a) the circumferential and longitudinal stress; (b) the circumferential and longitudinal strains; (c) the maximum in-plane and out-of-plane shear stresses; and (d) the normal and shear stresses acting on plane parallel to the welded seam. (20%)



【清大動械】

【中文翻譯】

壓力鋼桶以螺旋方式由與長度方向軸成 $\alpha = 60°$ 之捲繞焊成 (見圖)。桶具半徑 r=2 m ，壁厚 t=25mm，及內壓力 P=600kpa。此外鋼的彈性模數 $E = 200\ \text{GPa}$ ，而蒲松比 $\nu = 0.30$。求桶圓柱的部份之如下的量：(a) 圓周及長度方向應力，(b) 圓周及長度方向的應變(c) 最大平面上及出平面剪應力 (d) 作用於平行與垂直焊道平面上之正向與剪應力 (將這些應力繪示於正確指向的應力元素上。)

【解析】

(a)

$$\sigma_x = \frac{pr}{2t} = \frac{(6\times10^5)(2000)}{2(25)} = 24\times10^6\ \text{Pa}$$

$$= 24\ \text{MPa}$$

$$\sigma_y = \frac{pr}{t} = \frac{(6\times10^5)(2000)}{(25)} = 48\times10^6\ \text{Pa} = 48\ \text{MPa}$$

(b)

$$\varepsilon_x = \frac{\sigma_x}{E} - \nu\frac{\sigma_y}{E} = \frac{24\times10^6}{200\times10^9} - 0.3\times\frac{48\times10^6}{200\times10^9} = 4.8\times10^{-5}$$

$$\varepsilon_y = \frac{\sigma_y}{E} - \nu\frac{\sigma_x}{E} = \frac{48\times10^6}{200\times10^9} - 0.3\times\frac{24\times10^6}{200\times10^9} = 2.04\times10^{-4}$$

(c)

$$(\tau_{\max})_{\text{in-plane}} = \frac{\sigma_y - \sigma_x}{2} = 12\,\text{MPa}$$

$$(\tau_{\max})_{\text{out-of-plane}} = \frac{\sigma_y + p}{2} = 24.3\,\text{MPa}$$

(d)

$$\sigma = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 60^\circ = 30\,\text{MPa}$$

$$\tau = -\frac{\sigma_x - \sigma_y}{2}\sin 60^\circ = 10.39\,\text{MPa}$$

範例 3

(20%) A thin-walled cylindrical pressure vessel of radius r and thickness t is subjected simultaneously to internal gas pressure p and a compressive force F acting at the ends (see Figure 1). Assume $r \gg t$.

(a) (10%) What should be the magnitude of compressive force F in order to produce pure shear in the wall of the cylinder?

(b) (10%) If force $F = 60\pi$ kN, internal pressure p =12 MPa, inner radius r = 100 mm, and allowable normal and shear stresses are 100 MPa and 48 MPa, respectively, what is the required thickness of the vessel?



【105 台大應力】【104 成大醫工】

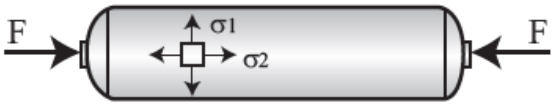
【中文翻譯】

薄圓柱壓力容器具內半徑為 r 、壁厚為 t 且受內部壓力 P ，同時也受一軸向負載 F 之壓縮(見圖)r>>t。(1) 軸向負載 F 為多少時可使薄圓柱壓力容器表面受到純剪應力(2)若軸向負載 $F = 60\pi$kN 、內部壓力 P=12MPa、壓力容器具內半徑為 r =100mm、最大容許內部應力為100MPa，殼壁之容許剪應力為48MPa，試求所需壁厚為多少。

【解析】

(1)應力元素分析

$$\sigma_\theta(\text{徑向應力}) = \sigma_1 = \frac{pr}{t} \quad \sigma_2 = \text{軸向應力} = \frac{pr}{2t} - \frac{F}{A} = \frac{pr}{2t} - \frac{F}{2\pi rt}$$



欲產生純剪應力態

$$\sigma_1 + \sigma_2 = 0 \Rightarrow \frac{Pr}{t} + \frac{Pr}{2t} - \frac{F}{2\pi rt} \Rightarrow F = 3\pi r^2 P$$

(2)求最大剪應力

$$\sigma_1 = \frac{Pr}{t} = \frac{12 \times 100}{t} = \frac{1200}{t}$$

$$\sigma_2 = \frac{Pr}{2t} - \frac{F}{2\pi rt} = \frac{300}{t}$$

$$(\tau_{max})_{\text{in plane}} = \frac{\sigma_1 - \sigma_2}{2} = \frac{450}{t} \le 48 \Rightarrow t \ge 9.375mm$$

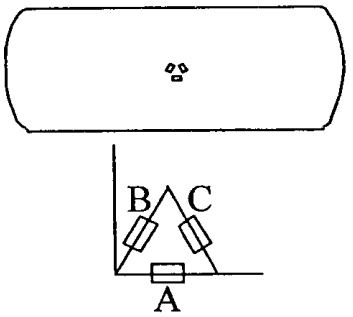
$$(\tau_{max})_{abs} = \frac{\sigma_1 - 0}{2} = \frac{600}{t} \le 48 \Rightarrow t \ge 12.5mm$$

$$\sigma_1 = \frac{1200}{t} \le 100 \Rightarrow t \ge 12mm$$

故選擇 $t \ge 12.5mm$

範例 4

A 60° strain-gage rosette (or delta rosette) is mounted on the outside of a cylindrical compressed air tank (see figure). The recorded strains are $\varepsilon_A = 40\times10^{-6}$ and $\varepsilon_B = \varepsilon_C = 137.5\times10^{-6}$. If the diameter of the tank is 1 m, the thickness of the wall is 40 mm, the modulus of elasticity is 200 GPa, and the Poisson's ratio is 0.3, what is the air pressure in the tank? (15%)



【100 清大動機】

【中文翻譯】

一 60° 應變規簇（ 或三角簇 ）貼於圓柱形壓縮空氣桶外表（ 見圖 ）。記錄的應變為 $\varepsilon_a = 40\times10^{-6}$ 及 $\varepsilon_b = \varepsilon_c = 137.5\times10^{-6}$。如果桶子之直徑為 1m 厚度 40mm，且彈性模數為 200 GPa 及 $\nu = 0.30$，桶的空氣壓力 p 為何？

【解析】

$$\sigma_x = \frac{PR}{2t} = \frac{P\times500}{2\times40} = 6.25P$$

$$\sigma_y = \frac{PR}{t} = \frac{P\times500}{40} = 12.5P \text{ 、 } \sigma_z = 0$$

$$\varepsilon_x = \frac{1}{E}\times[\sigma_x - \upsilon(\sigma_y + \sigma_z)] = \frac{1}{200\times10^3}\times[6.25P - 0.3(12.5P)] = 12.5P\times10^{-6}$$

$$\varepsilon_y = \frac{1}{E}\times[\sigma_y - \upsilon(\sigma_x + \sigma_z)] = \frac{1}{200\times10^3}\times[12.5P - 0.3(6.25P)] = 53.125P\times10^{-6}$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G} = 0$$

$$\varepsilon_A = 40\mu \Rightarrow 12.5P = 40$$

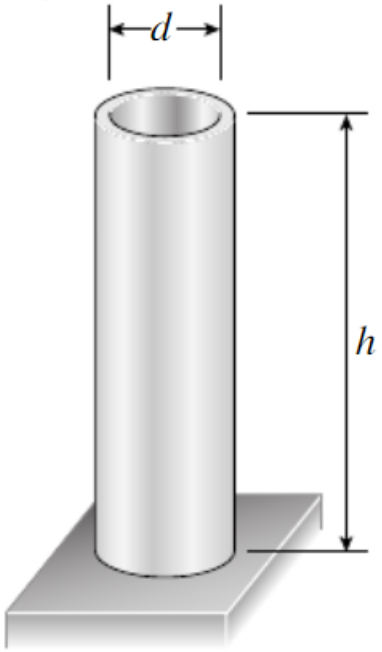
$$\Rightarrow P = 3.2(MPa)$$

範例 5

A tall standpipe with an open top (see figure) has diameter $d = 2.2$ m and wall thickness $t = 20$ mm.

(a) What height h of water will produce a circumferential stress of 12 MPa in the wall of the standpipe?

(b) What is the axial stress in the wall of the tank due to the water pressure?



【105 台大工科】

【中文翻譯】

一立柱如圖具開上頂，直徑 $d = 2.2$m 及壁厚 $t = 20$mm 。(a) 水高多少時壁會產生圓周應力 12 MPa？(b) 由於水壓，壁之軸向應力為何？

【解析】

(a)

$d = 2.2$ m $\quad r = 1.1$ m $\quad t = 20$ mm

$\gamma = 9.81$ kN/m^3

$p = \gamma h$

$$\sigma_1 = \frac{pr}{t} = 12 \text{ MPa} = \frac{0.00981h\,(1.1 \text{ m})}{20 \text{ mm}}$$

$$h = \frac{12\,(20)}{0.00981\,(1.1)} = 22.2 \text{ m}$$

(b)

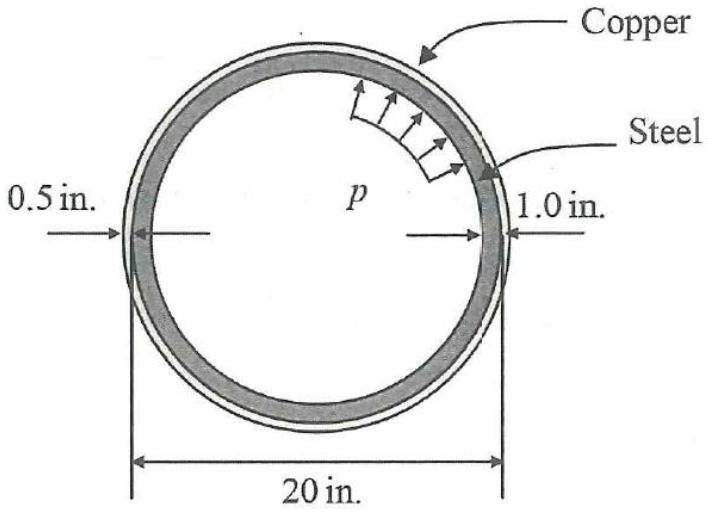
由於為開口薄閉管

故軸向方向之應力=0

範例 6

A thin-walled copper tube is fitted just snugly over a steel tube at 420˚F. The fit is such that no pressure exists between the tubes. The assembly is then cooled down to 70˚F and subjected to a uniform inner pressure P=50psi. Calculate the radial pressure q between the tubes and the stress in the copper tube. To simplify the calculations, assume that each tube has a mean radius r=10in.

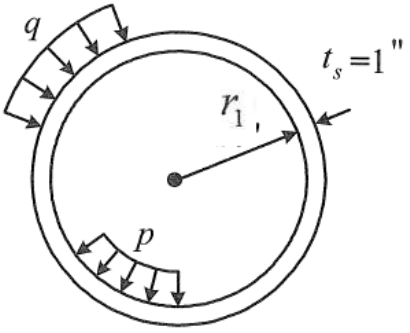
$E_s = 30\times10^6$ psi $\quad E_c = 13\times10^6$ psi $\quad \alpha_s = 6.5\times10^{-6}/^\circ\text{F}$ $\quad \alpha_c = 10.4\times10^{-6}/^\circ\text{F}$



【清大動機】

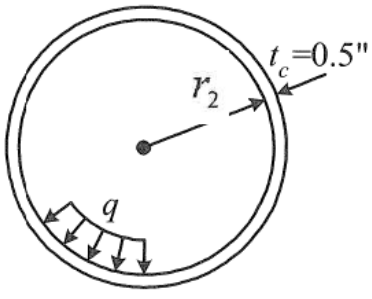
【解析】

鋼的內徑 $r_1 = 9$ in.



內鋼管

鋼的外徑 $r_2 = 10$ in.



外銅管

$$\sigma_C = \frac{qr_2}{t_C} \qquad \varepsilon_C = \frac{\sigma_C}{E_C} + \alpha_C \Delta T = \frac{qr_2}{E_C t_C} + \alpha_C \Delta T$$

$$\sigma_S = \frac{pr_1}{t_S} - \frac{qr_2}{t_S} \qquad \varepsilon_S = \frac{\sigma_S}{E_S} + \alpha_S \Delta T = \frac{pr_1}{E_S t_S} - \frac{qr_2}{E_S t_S} + \alpha_S \Delta T$$

相合方程式 $\varepsilon_C = \varepsilon_S$

$$\frac{qr_2}{E_C t_C} + \alpha_C \Delta T = \frac{pr_1}{E_S t_S} - \frac{qr_2}{E_S t_S} + \alpha_S \Delta T$$

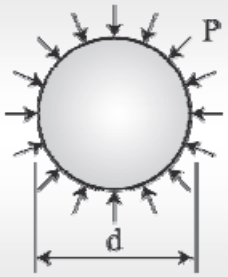
代入得 $q = 737.26$ psi

$$\sigma_C = \frac{qr_2}{t_C} = 14745 \text{ psi} \quad \text{(拉應力)}$$

$$\sigma_S = \frac{pr_1}{t_S} - \frac{qr_2}{t_S} = -6923 \text{ psi} \quad \text{(壓應力)}$$

範例 7

如圖所示，一實心球體直徑為 10cm，**承受均勻壓力** P＝10,000Pa，**假設此球體楊氏係數** $E=30\times10^6$ Pa，**蒲松比** $\nu=0.3$。**試求此球體之體積減少量**。(94 年機械高考)



【解析】

解 由體積虎克定律

$$\varepsilon_\forall = \frac{\Delta\forall}{\forall} = \varepsilon_x + \varepsilon_y + \varepsilon_z = \frac{3(1-2\nu)}{E}\left[\frac{\sigma_x+\sigma_y+\sigma_z}{3}\right] = \frac{\sigma_m}{k}$$

其中 $k = \dfrac{E}{3(1-2\nu)} = \dfrac{30\times10^6}{3(1-2\times0.3)} = 25\times10^6$ (Pa)

球體積 $\forall = \dfrac{\pi}{6}d^3 = \dfrac{\pi}{6}\times(10\times10^{-2})^3 = 5.236\times10^{-4}$

又圓球承受均勻壓力 $P = 10^4$ Pa $= \sigma_m$(壓)

$$\varepsilon_\forall = \frac{\Delta\forall}{\forall} = \frac{\Delta\forall}{5.236\times10^{-6}} = \frac{(-10^4)}{25\times10^6} \Rightarrow \Delta\forall = -2.09\times10^{-7}(m^3)$$

體積減少量為 2.09×10^{-7} m^3

範例 8

一個球形鋼製壓力容器(spherical steel pressure vessel)的半徑為 10 cm、厚度為 2 mm。鋼的楊氏模數(Young's modulus) E 為 200 GPa，柏松比(Poisson's ratio) ν 為 0.3。請問如果在容器內加壓至 10 atm，其半徑會增加的比例為 (A) 0.0075% (B) 0.00875% (C) 0.005% (D) 0.01% (E)以上皆非

承上題，不過容器改為一個封閉圓柱形的鋼製容器(closed-ended cylindrical steel pressure vessel)，半徑為 10 cm、厚度為 2 mm，長度為 1 m，同樣的材料。請問如果在容器內加壓至 10 atm，其長度會增加的比例為 (A) 0.015% (B) 0.05% (C) 0.005% (D) 0.0015% (E)以上皆非

【101 台大機械】

【解析】

(一)

$$\sigma_x = \sigma_y = \frac{PR}{2t} = \frac{10\times10^5\times100}{2\times2} = 250\times10^5(Pa) = 25(MPa)$$

$$\varepsilon_x = \frac{1}{E}\times[\sigma_x - \upsilon(\sigma_y)] = \frac{1}{200\times10^3}\times[25-0.3(25)] = 87.5\times10^{-6} = 0.00875\%$$

故選B

(二)

$$\sigma_x = \frac{PR}{2t} = \frac{10\times10^5\times100}{2\times2} = 250\times10^5(Pa) = 25(MPa)$$

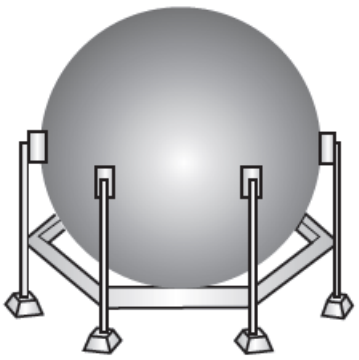
$$\sigma_y = \frac{PR}{t} = \frac{10\times10^5\times100}{2} = 500\times10^5(Pa) = 50(MPa)$$

$$\varepsilon_x = \frac{1}{E}\times[\sigma_x - \upsilon(\sigma_y)] = \frac{1}{200\times10^3}\times[25-0.3(50)] = 50\times10^{-6} = 0.005\%$$

故選C

範例 9

一球型氣體儲槽，具內半徑r＝1.5m。若其承受內壓力p＝300kPa，且正向應力不超過12MPa，求其所需厚度。

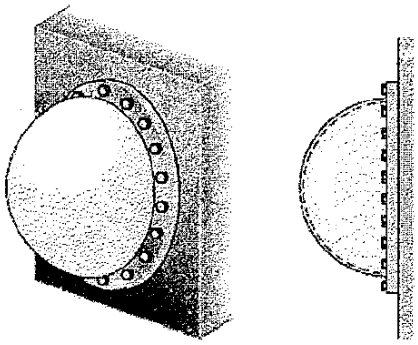


【解析】

$$\sigma_{allow}=\frac{pr}{2t}\ ;\ 12\left(10^{6}\right)=\frac{300\left(10^{3}\right)(1.5)}{2t} \qquad t=0.0188\,m=18.8\,mm$$

範例 10

5. A hemispherical window (or *viewport*) in a decompression chamber (see figure) is subjected to an internal air pressure of 600 kPa. The port is attached to the wall of the chamber by 18 bolts. Find the tensile force F in each bolt and the tensile stress σ in the viewport if the radius of the hemisphere is 400 mm and its thickness is 25 mm. (15%)

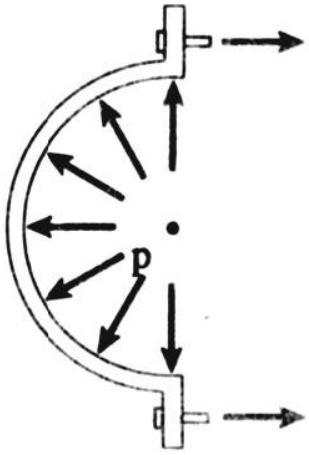


【105 成大醫工】

【中文翻譯】

減壓室半球窗（ 或視窗)(見圖 ）受內壓力600 kPa。接面以18支栓接合於牆壁。如果半球半徑為400 mm，且厚度為25 mm，求每支螺栓的拉伸力 F 及視窗的拉伸應力 σ 。

【解析】



$$T = pA = p(\pi r^2)$$

$$F = \frac{p(\pi r^2)}{18} = \frac{(600)(\pi)(0.4)}{18} = 16.76\ \text{kN}$$

$$\sigma = \frac{pr}{2t} = \frac{(600)(0.4)}{2(10.25\ \text{m})} = 4800\ \text{kPa}$$

實戰演練

例 1

As shown in **Fig. 4**, the cylindrical tank is fabricated by welding a strip of thin plate helically and making an angle θ with the longitudinal axis of the tank. The strip has a width w and thickness t. The gas within the tank of diameter d is pressured to p. Please determine the normal stress developed along the strip. **(25%)**

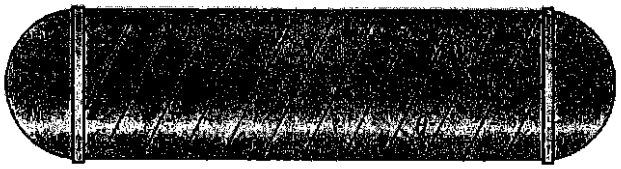


Fig. 4

【104 中央機械】

【解析】

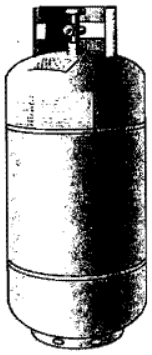
$$\sigma_x = \frac{PR}{2t} = \frac{Pd}{4t}$$

$$\sigma_y = \frac{PR}{t} = \frac{Pd}{2t}$$

$$\sigma_\theta = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos(2\times\theta) + \tau_{xy}\sin 2\theta = \frac{3Pd}{8t} + \frac{Pd}{8t}\cos 2\theta$$

例 2

(10pts) The bulk storage tank shown has as outer diameter of 3.3 m and a wall thickness of 18 mm. At a time when the inter pressure of the tank is 1.5 MPa, determine the maximum normal stress and the maximum shearing stress in the tank.



【101 成大工科】

【解析】

$$\sigma_x = \frac{PR}{t} = \frac{1.5 \times 1650}{18} = 137.5(MPa)$$

$$\sigma_y = \frac{PR}{2t} = \frac{1.5 \times 1650}{2 \times 18} = 68.75(MPa) \text{ 、 } \sigma_z = -1.5(MPa)$$

$$\sigma_1 = \sigma_x = 137.5(MPa) \text{ 、 } \sigma_2 = 68.75(MPa) \text{ 、 } \sigma_3 = -1.5(MPa)$$

$$\tau_{\max} = \frac{\sigma_1 - \sigma_2}{2} = 69.5(MPa)$$

例 3

一鋼製之圓筒形薄壁壓力容器，內徑為 150 mm，承受內壓力 p = 12 MPa。已知鋼材之拉伸降伏應力為 300 MPa，剪降伏強度為 140 MPa。設對降伏之安全因數為 2.0，試筒壁所需之最小厚度。

【解析】

(1)考慮拉伸容許應力

$$\sigma_1 = \frac{pd}{2t} \text{，} \frac{300}{2.0} = \frac{(12)(150)}{2t} \text{，} t = 6.00 \text{ mm}$$

(2)考慮容許剪應力

$$\tau_{max} = \frac{pd}{4t} \text{，} \frac{140}{2.0} = \frac{12(150)}{4t} \text{，} t = 6.43 \text{ mm}$$

故所需最小厚度 $t = 6.43$ mm (取較大者)

例 4

The maximum gage pressure is known to be 8 MPa in a spherical steel pressure vessel having a 250-mm outer diameter and a 6-mm wall thickness. Knowing that the ultimate stress in the steel used is $\sigma_U = 400$ MPa, determine the factor of safety with respect to tensile failure.

【解析】

$$r = \frac{d}{2} - t = \frac{250}{2} - 6 = 119 \text{ mm} = 119 \times 10^{-3} \text{ m}, \quad t = 6 \times 10^{-3} \text{ m}$$

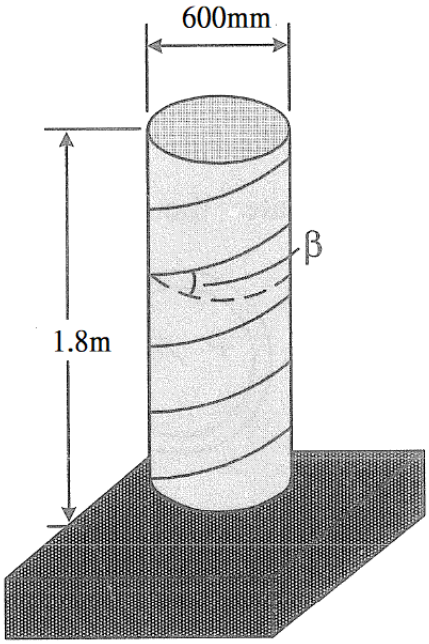
$$\sigma_1 = \sigma_2 = \frac{pr}{2t} = \frac{(8 \times 10^6 \text{ Pa})(119 \times 10^{-3} \text{ m})}{2(6 \times 10^{-3} \text{ m})} = 79.333 \times 10^6 \text{ Pa}$$

$$F.S. = \frac{\sigma_U}{\sigma_{\max}} = \frac{400 \times 10^6}{79.333 \times 10^6}$$

$$F.S. = 5.04$$

例 5

The cylindrical portion of the compressed-air tank shown is fabricated of 8-mm-thick plate welded along a helix forming an angle $\beta = 30^\circ$ with the horizontal. Knowing that the allowable stress normal to the weld is 75 MPa, (a)determine the largest gage pressure that can be used in the tank.(b)Determine the gage pressure that will cause a shearing stress parallel to the weld of 30 MPa.



【成大工科】【105 台大機械類題】

【解析】

$$\sigma_x = \frac{pr}{t}$$

$$\sigma_y = \frac{1}{2}\frac{pr}{t}$$

$$\sigma_{ave} = \frac{1}{2}(\sigma_x + \sigma_y) = \frac{3}{4}\frac{pr}{t}$$

$$R = \frac{\sigma_x - \sigma_y}{2} = \frac{1}{4}\frac{pr}{t}$$

$$r = \frac{1}{2}d - t = \frac{1}{2}(600) - 8 = 292 \text{ mm}$$

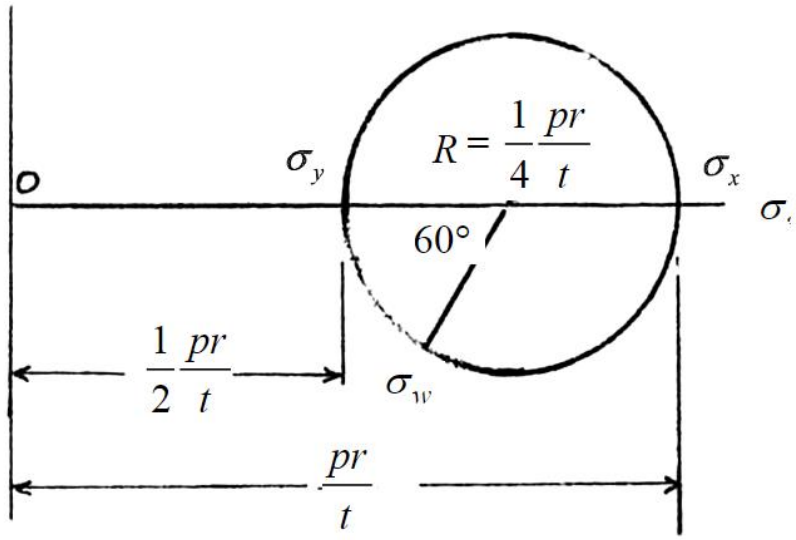
$$\sigma_w = \sigma_{\text{ave}} - R\cos 60°$$

$$= \frac{5}{8}\frac{pr}{t}$$

$$p = \frac{8}{5}\frac{\sigma_w t}{r} = \frac{8}{5}\frac{(75)(8)}{292} = 3.29 \text{ MPa}$$

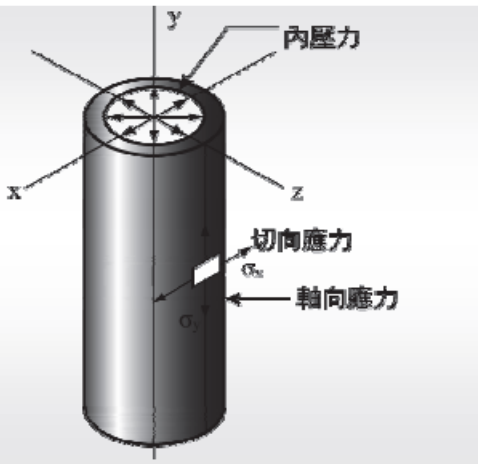
$$\tau_w = R\sin 60° = \frac{\sqrt{3}}{8}\frac{pr}{t}$$

$$p = \frac{8}{\sqrt{3}}\frac{\tau_w t}{R} = \frac{8}{\sqrt{3}}\frac{(30)(8)}{292} = 3.80 \text{ MPa}$$



例 6

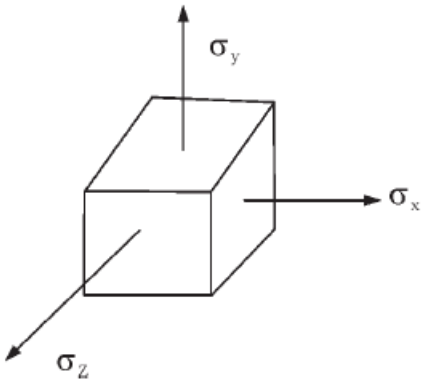
薄管壓力容器如圖所示，內部之蒸汽壓力 P=500psi，其中容器之內徑 D=4in，容器厚度 t=0.08in，容器兩端是封閉的。如此典型的機械元件，所承受的是三維的應力狀態，試分析最大剪應力及其對應的方位。



【機械高考】

【解析】

解 1. 取三維應力元素：



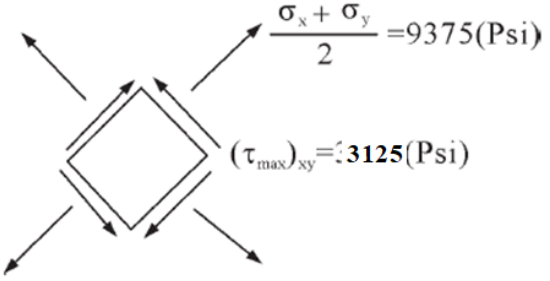
$$\sigma_x = \frac{PR}{t} = \frac{500\times\frac{4}{2}}{0.08} = 12500(psi) \quad \sigma_y = \frac{PR}{2t} = \frac{500\times 2}{2\times 0.08} = 6250(psi)$$

因此主平面應力 $\begin{cases}\sigma_1 = \sigma_x = 12500psi \\ \sigma_2 = \sigma_y = 6250psi\end{cases}$

2. **最大平面上剪應力：**

$$(\tau_{max})_{xy} = \frac{\sigma_1 - \sigma_2}{2} = \frac{12500 - 6250}{2} = \mathbf{3125}$$

平面應力元素圖如下所示



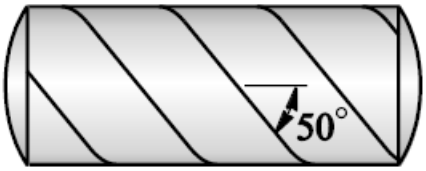
3. **絕對最大剪應力：**

$\sigma_1 > \sigma_2 > \sigma_3$ 其中 $\sigma_3 = \sigma_z = 0$

$$\tau_{max} = \frac{\sigma_1 - \sigma_3}{2} = \frac{12500 - 0}{2} = 6250\text{psi}$$

例 7

圖中筒狀薄壁壓力容器之內徑為 500 mm，壁厚為 8 mm，內壓力為 1.5 MPa。圓筒上之焊道與縱向軸線之夾角為 50°，試求(a)周向應力σ_h與縱向應力σ_a，(b)同平面最大剪應力，(c)絕對最大剪應力，(d)焊道所受之正交應力與剪應力。



【清大動機類題】

【解析】

$d = 500$ mm，$t = 8$ mm，$p = 1.5$ MPa，

(a) $\sigma_h = \dfrac{pd}{2t} = \dfrac{(1.5)(500)}{2(8)} = 46.88$ MPa

$\sigma_a = \dfrac{pd}{4t} = 23.44$ MPa

(b) $\tau_{\max} = \dfrac{1}{2}(\sigma_h - \sigma_a) = \dfrac{1}{2}(46.88 - 23.44) = 11.72$ MPa

(c) $(\tau_{\max})_{abs} = \dfrac{\sigma_h}{2} = \dfrac{pd}{4t} = 23.44$ MPa

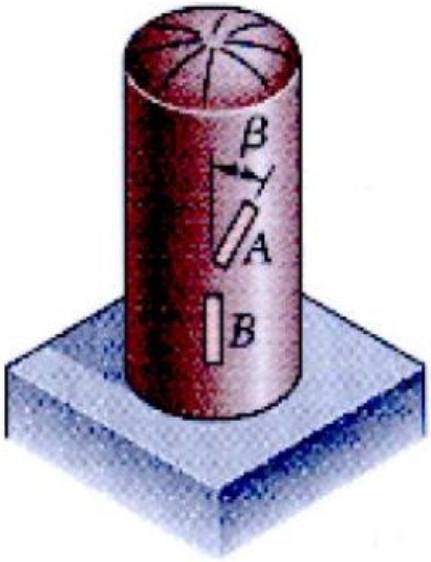
(d) $\sigma_x = \sigma_a = 23.44$ MPa，$\sigma_y = \sigma_h = 46.88$ MPa，$\theta = 40°$

$$\sigma_{x'} = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2}\cos 2\theta = \frac{23.44 + 46.88}{2} + \frac{23.44 - 46.88}{2}\cos 80° = 33.1 \text{ MPa}$$

$$\tau_{x'y'} = -\frac{\sigma_x - \sigma_y}{2}\sin 2\theta = -\frac{23.44 - 46.88}{2}\sin 80° = 11.54 \text{ MPa}$$

例 8

(15pts) A single strain gage *A* forming an angle $\beta = 30°$ with the vertical is used to determine the gage pressure in the cylindrical steel tank shown. The cylindrical wall of the tank is 10 mm thick, has a 900 mm inner diameter, and is made of a steel with E = 200 GPa and $\nu = 0.30$. Determine the pressure in the tank corresponding to a gage reading of 220×10^{-6} mm/ mm ($\cos 30° = 0.866$).



【104 成大工科 】【105 成大工科】 【107 成大工科】

【解析】

$$\sigma_x = \frac{PR}{t} = \frac{P \times 450}{10} = 45P$$

$$\sigma_y = \frac{PR}{2t} = 22.5P \text{ 、 } \sigma_z = 0$$

$$\varepsilon_x = \frac{1}{E} \times [\sigma_x - \upsilon(\sigma_y + \sigma_z)] = \frac{1}{200 \times 10^3} \times [45P - 0.3(22.5P)] = 191.25P \times 10^{-6}$$

$$\varepsilon_y = \frac{1}{E} \times [\sigma_y - \upsilon(\sigma_x + \sigma_z)] = \frac{1}{200 \times 10^3} \times [22.5P - 0.3(45P)] = 45P \times 10^{-6}$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G} = 0$$

$$\varepsilon_A = 40\mu \Rightarrow 12.5P = 40$$

$$\Rightarrow P = 2.7(MPa)$$

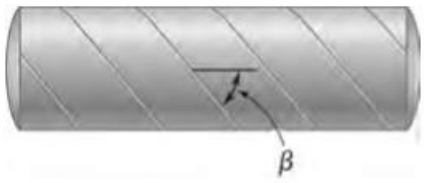
$$\varepsilon_{60°} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2} \cos 2\theta + \frac{\gamma_{xy}}{2} \sin 2\theta = 220\mu$$

$$\Rightarrow \frac{191.25P + 45P}{2} + \frac{191.25P - 45P}{2} \cos 2 \times (60^\circ) = 220\mu$$

$\Rightarrow P = 2.7(MPa)$

例 9

The steel pressure tank shown has a 750-mm inner diameter and a 9-mm wall thickness. Knowing that the butt-welded seams form an angle $\beta = 50°$ with the longitudinal axis of the tank and that the gage pressure in the tank is 1.5 MPa, determine (*a*) the normal stress perpendicular to the weld, (*b*) the shearing stress parallel to the weld.



【解析】

$$r = \frac{d}{2} = 375\text{ mm} = 0.375\text{ m}$$

$$\sigma_1 = \frac{pr}{t} = \frac{(1.5\times10^6\text{ Pa}\times0.375\text{ m})}{0.009\text{ m}} = 62.5\times10^6\text{ Pa} = 62.5\text{ MPa}$$

$$\sigma_2 = \frac{1}{2}\sigma_1 = 31.25\text{ MPa} \qquad 2\beta = 100°$$

$$\sigma_{ave} = \frac{1}{2}(\sigma_1 + \sigma_2) = 46.875\text{ MPa}$$

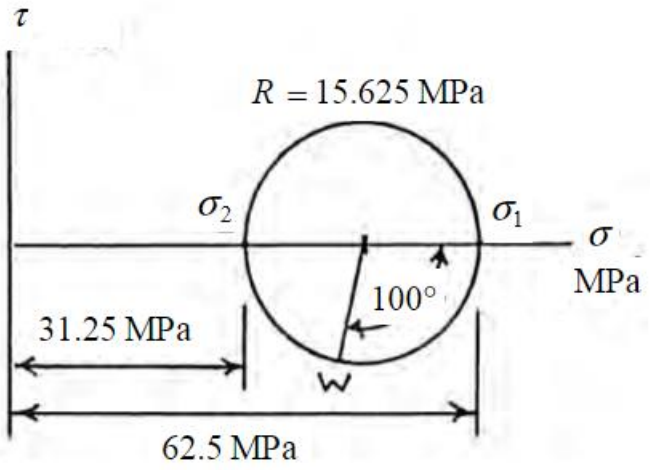
$$R = \frac{\sigma_1 - \sigma_2}{2} = 15.625\text{ MPa}$$

$$\sigma_w = \sigma_{ave} + R\cos100°$$

$$\sigma_w = 44.2\text{ MPa}$$

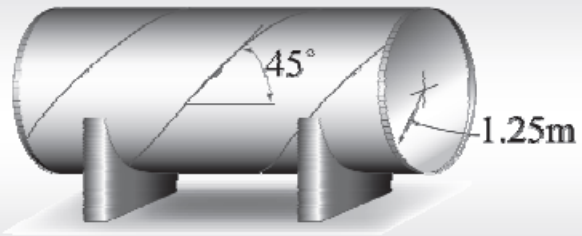
$$\tau_w = R\sin100°$$

$$\tau_w = 15.39\text{ MPa}$$


τ
R = 15.625 MPa
σ2
σ1
σ
MPa
100°
31.25 MPa
W
62.5 MPa

例 10

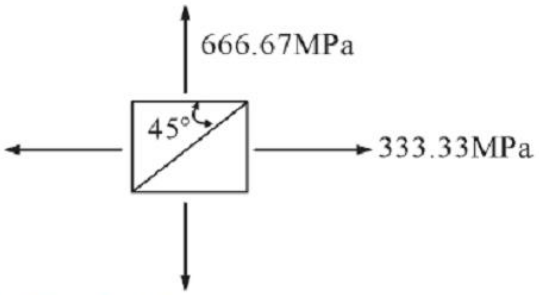
圓筒型壓力容器如圖所示，容器內半徑 1.25m，承受壓力 8MPa，壁厚 15mm，其中鋼板沿與水平成 45°焊縫，求接縫所受正向應力及剪應力。



【解析】

解 1. $\sigma_x = \frac{Pr}{2t} = \frac{8\times(1.25)}{2(0.015)} = 333.33(MPa)$

$\sigma_y = \frac{Pr}{t} = \frac{8\times(1.25)}{0.015} = 666.67(MPa)$



2. 焊縫與水平呈 45°：

$$\sigma_{45°} = \frac{\sigma_x+\sigma_y}{2} + \frac{\sigma_x-\sigma_y}{2}\cos(2\times -45°) + \tau_{xy}\sin(2\times -45°)$$

$$= \frac{333.33+666.67}{2} = 500(MPa)$$

$$\tau_{45°} = -\left(\frac{\sigma_x-\sigma_y}{2}\right)\sin(2\times -45°) + \tau_{xy}\cos(2\times -45°)$$

$$= -\left(\frac{333.33-666.67}{2}\right) = 166.67(MPa)$$

正向應力 = 500(MPa)　剪應力 = 166.67(MPa)

例 11

Let E and ν be the Young's modulus and Poisson's ratio of the material. Both are assumed to be onstant.

(a).(5%). Let(x, y) denote the conventional rectangular coordinate. Let σ_x and σ_y be the normal stresses on planes perpendicular to the x amd y axes, and ε_x and ε_y in terms of σ_x, σ_y, E and ν.

(b).(5%). Consider a thin-walled circullar cylindrial tank with closed ends and an internal pressure p. Let t and r be the thickness of the wall and the inside radius of the cylinder, respectively. Assmed $r >> t$. Let σ_α be the stress in the axial direction and σ_k be the stress in the circumferential direction. Find σ_α and σ_k.

(c).(5%). Let L be the axial length of the above tank. Find the changes of the axial length L and the inner radius r of the tank caused by the internal pressure p.

【台大應力】

【解析】

(a)

$$\varepsilon_x = \frac{1}{E}(\sigma_x - \nu\sigma_y) \ ; \ \varepsilon_y = \frac{1}{E}(\sigma_y - \nu\sigma_x)$$

(b)

$$\sigma_\alpha = \frac{p \cdot L}{2t}$$

$$\sigma_k = \frac{p \cdot L}{t}$$

(c)

$$\varepsilon_a = \frac{1}{E}(\sigma_a - \nu\sigma_h) = \frac{pr}{2Et}(1-2\nu)$$

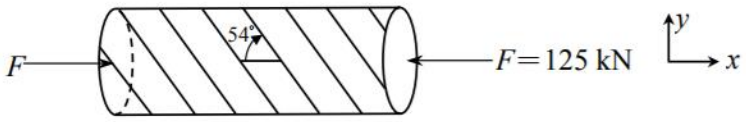
$$\Delta_a = (\varepsilon_a)(L) = \frac{prL}{2Et}(1-2\nu)$$

$$\varepsilon_h = \frac{1}{E}(\sigma_h - \nu\sigma_a) = \frac{pr}{2Et}(2-\nu)$$

$$\Delta_r = (\varepsilon_h)(r) = \frac{pr^2}{2Et}(2-\nu)$$

例 12

、設有一鋼製圓筒容器的內半徑r_i為 600 mm，厚度是 16 mm。該容器內部承受 1750 kPa錶壓力，同時外部承受 125 kN的軸向力，如圖一所示。該容器是以端焊（butt weld）接合方式構造，焊接道相對於軸向的角度是 54°。㈠試繪出焊道處容器表面元素（element）的應力狀態（state of stress）圖。㈡試求垂直於焊接道的應力。㈢試求平行於焊接道的剪應力。（提示：薄壁壓力容器的箍應力（hoop stress）為 $\sigma_h = \frac{pr_i}{t}$，軸向應力（axial stress）為 $\sigma_a = \frac{pr_i}{2t}$）（20 分）



【土木地特三等】

【解析】

$\sigma_x = \frac{pr}{2t} - \frac{F}{A}$; $\sigma_y = \frac{pr}{t}$　　內壓力p造成軸向張力

$$\sigma_x(MPa) = \frac{1750\times 600}{2\times 16}\cdot 10^{-3} - \frac{125\cdot 10^3}{\pi\cdot\left(616^2-600^2\right)} = 32.81 - 2.05 = 30.76MPa$$

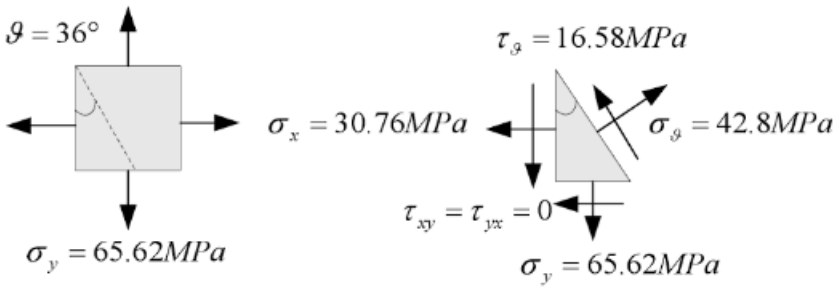
$$\sigma_y(MPa) = \frac{1750\times 600}{16}\cdot 10^{-3} = 65.62MPa$$

$$\sigma_\theta = \left(\frac{\sigma_x+\sigma_y}{2}\right) + \left(\frac{\sigma_x-\sigma_y}{2}\right)\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\sigma_{(\theta=36°)} = \left(\frac{30.77+65.63}{2}\right) + \left(\frac{30.77-65.63}{2}\right)\cos(2\times 36°) + 0$$
$$= 42.8\,\text{MPa}$$

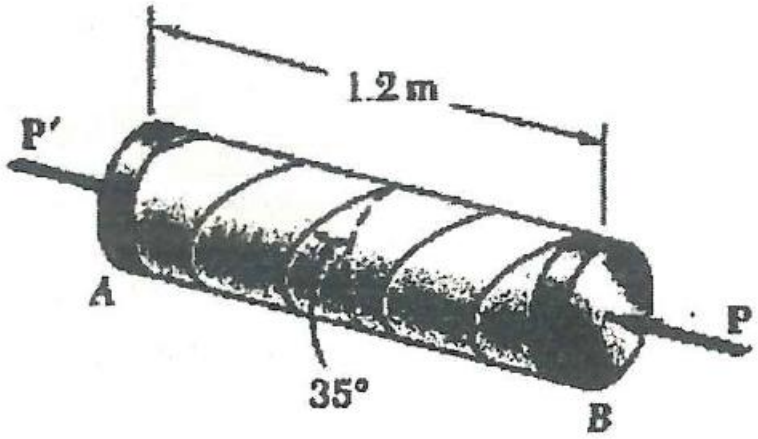
$$\tau_\theta = -\left(\frac{\sigma_x-\sigma_y}{2}\right)\sin 2\theta + \tau_{xy}\cos 2\theta$$

$$\tau_{(\theta=36°)} = -\left(\frac{30.77-65.63}{2}\right)\sin(2\times 36°) + 0 = 16.58\,\text{MPa}$$


$\vartheta = 36°$
$\sigma_x = 30.76MPa$
$\sigma_y = 65.62MPa$
$\tau_\vartheta = 16.58MPa$
$\sigma_\vartheta = 42.8MPa$
$\tau_{xy} = \tau_{yx} = 0$
$\sigma_y = 65.62MPa$

例 13

(20pts) A pressure vessel of 250-mm inner diameter and 6-mm wall thickness is fabricated from a 1.2-m section of spirally welded pipe AB and is equipped with two rigid end plates. The gage pressure inside the vessel is 2 MPa and 45-kN centric axial force P and P' are applied to the end plates. Determine (a) the normal stress perpendicular to the weld, (b) the shear stress parallel to the weld.



【102 成大工科】【106 成大工科】

【解析】

$$A=\frac{\pi}{4}\left(262^2-250^2\right)=4.825\times10^3\ \text{mm}^2$$

$$\sigma_x=\frac{pr}{2t}-\frac{P}{A}=\frac{2\times125}{2\times6}-\frac{45000}{4.825\times10^3}\ \text{MPa}=11.507\ \text{MPa}$$

$$\sigma_y=\frac{pr}{t}=\frac{2\times125}{6}\ \text{MPa}=41.667\ \text{MPa}\qquad \tau_{xy}=0$$

$$\sigma_{:\theta}=\frac{\sigma_x+\sigma_y}{2}+\frac{\sigma_x-\sigma_y}{2}\cos2\theta+\tau_{xy}\sin2\theta=21.43\ \text{MPa}$$

$$\tau_{:\theta}=-\frac{\sigma_x-\sigma_y}{2}\sin2\theta+\tau_{xy}\cos2\theta=-14.17\ \text{MPa}$$

例 14

A solid sphere of diameter 200 mm is subjected to an external pressure of 100 kPa. The sphere has modulus of elasticity $E = 1$ GPa and Poisson's ratio $\nu = 0.25$, find the change in diameter of the sphere. A rubber ball is inflated to a pressure of 100 kPa. At that pressure the diameter is 200 mm and the wall thickness is 2 mm. The rubber has modulus of elasticity E = 5 MPa and Poisson's ratio $\nu = 0.5$. Determine the maximum stress and strain in the ball.

【清大動機】

【解析】

$$\varepsilon = \frac{1-2\nu}{E}(-p) = \frac{1-0.5}{1000}(-0.1) = -5\times10^{-5}$$

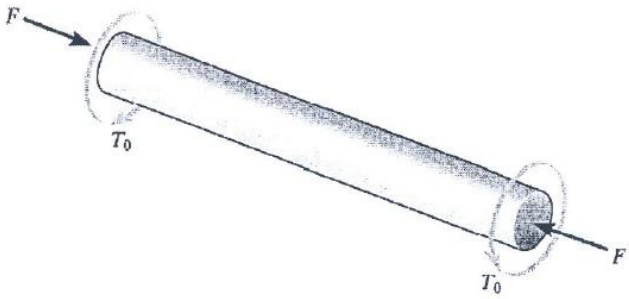
$$\Rightarrow \Delta d = \varepsilon \times d = (-5\times10^{-5})\times 200 = -0.01(\text{mm})$$

$$\sigma = \frac{pr}{2t} = \frac{0.1\times0.1}{2\times0.002} = 2.5(\text{MPa})$$

$$\varepsilon = \frac{1}{E}(2.5 - 0.5\times2.5) = 0.25$$

例 15

6. The solid shaft has a diameter d and is subjected to the loadings as shown below.



Please:

(a) Determine the normal stress and shear stress on the surface of the shaft. [10pts]

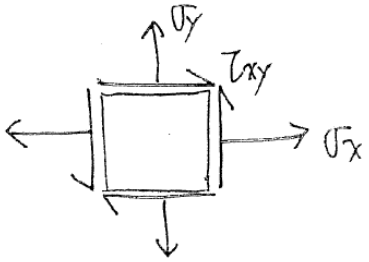
(b) Draw Mohr's circle. [5pts]

(c) Determine the principle stresses and maximum in-plane shear stress that is developed anywhere on the surface of the shaft. [10pts]

【106 清大動機】

【解析】

<S01>: (11) 軸表面之材料點



$$\sigma_x = -\frac{F}{A} = -\frac{4F}{\pi d^2}$$

$$\sigma_y = 0$$

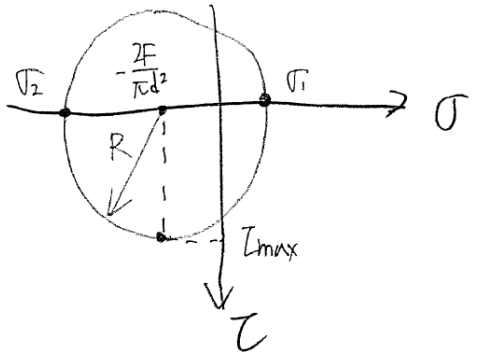
$$\tau_{xy} = -\frac{T_0 r}{J} = -\frac{16T_0}{\pi d^3}$$

(2) 莫耳圓

① 圓心 $= \left(\frac{\sigma_x+\sigma_y}{2}, 0\right) = \left(-\frac{2F}{\pi d^2}, 0\right)$

② 半徑 $= R = \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2 + \tau_{xy}^2}$

$= \frac{2}{\pi d^3}\sqrt{F^2d^2+64T_0^2}$



(3) 主應力及最大剪應力

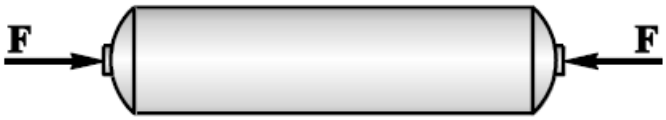
$\sigma_{1,2} = -\frac{2F}{\pi d^2} \pm \frac{2}{\pi d^3}\sqrt{F^2d^2+64T_0^2}$

$\tau_{max} = R$

$= \frac{2}{\pi d^3}\sqrt{F^2d^2+64T_0^2}$

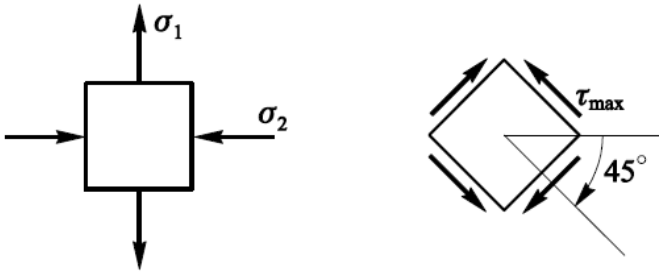
例 16

圖中內半徑為 r 之筒狀薄壁壓力容器，內壓力為 p，同時兩端承受壓力 F，若欲使筒壁承受純剪，則壓力 F 之大小為何？圓筒之斷面積 $A = 2\pi rt$。



【105 台大應力】

【解析】



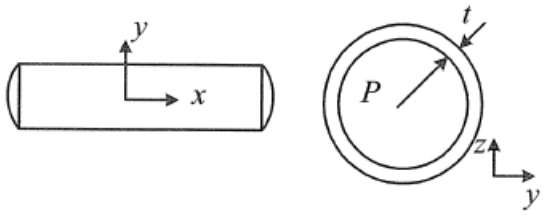
當雙軸向應力狀態之 $\sigma_1 = -\sigma_2$ 時，在 45°方位為純剪狀態，且 $\tau_{max} = \sigma_1$

$$\sigma_1 = \frac{pr}{t} \text{ ，} \sigma_2 = \frac{pr}{2t} - \frac{F}{2\pi rt}$$

$$\sigma_1 = -\sigma_2 \text{ ，} \frac{pr}{t} = -\frac{pr}{2t} + \frac{F}{2\pi rt} \text{ ，} F = 3\pi pr^2$$

例 17

The internal pressure (P) of a closed cylindrical vessel is 2 MPa. The radius (r) is 20 cm and the thickness (t) is 1 mm. The configuration is shown in Figure. (1) Calculate the circumferential stress (σ_{yy}) and longitudinal stress (σ_{xx}). (2) Calculate the principal stresses at lateral outer surface. (3) What is the absolute maximum shear stress at the lateral outer surface? (4) Calculate the principal stresses at lateral inner surface.



【台大應力】

【解析】

(1)

$$\sigma_{yy} = \frac{pr}{t} = 400(MPa)$$

$$\sigma_{xx} = \frac{pr}{2t} = 200(MPa)$$

(2)

$$\sigma_1 = 400MPa \quad , \quad \sigma_2 = 200MPa \quad , \quad \sigma_3 = 0$$

(3)

$$\tau_{\max} = \frac{\sigma_1 - \sigma_3}{2} = 200(MPa)$$

(4)

$$\sigma_1 = 400MPa \quad , \quad \sigma_2 = 200MPa \quad , \quad \sigma_3 = -2MPa$$

例 18

一微小應變規(strain gage)貼在一薄壁圓柱壓力容器表面量測應變(圖 13.1 與圖 13.2)。該容器直徑為 250 mm、壁厚為 5 mm、楊氏係數 $E=200$ GPa、泊松比 $\nu=0.25$、降伏應力 $\sigma_Y=30$ MPa。若量測的應變為 $\varepsilon_g=100\times10^{-6}$。

(1) 請問容器內的壓力為何？ (3%)

(2) 請畫出莫爾圓(Mohr's circle)求出主應力和最大剪應力絕對值，並請說明容器在此受力狀態是否會產生塑性變形？ (7%)

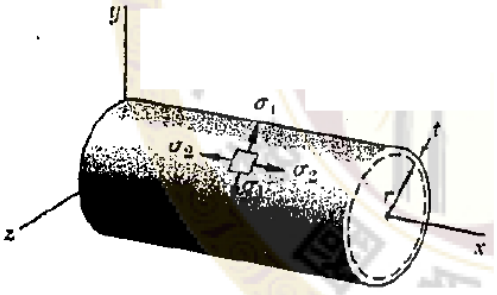


圖 13.1 薄壁壓力容器

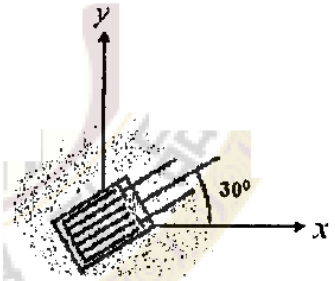


圖 13.2 應變規貼在容器表面的方向

【102 台大機械】

【解析】

(一)

$$\sigma_x=\frac{PR}{2t}=\frac{P\times125}{2\times5}=12.5P$$

$$\sigma_y=\frac{PR}{t}=\frac{P\times125}{2\times5}=25P\ 、\ \sigma_z=0$$

$$\varepsilon_x=\frac{1}{E}\times[\sigma_x-\upsilon(\sigma_y+\sigma_z)]=\frac{1}{200\times10^3}\times[12.5P-0.25(25P)]=31.25P\times10^{-6}$$

$$\varepsilon_y=\frac{1}{E}\times[\sigma_y-\upsilon(\sigma_x+\sigma_z)]=\frac{1}{200\times10^3}\times[25P-0.25(12.5P)]=109.375P\times10^{-6}$$

$$\gamma_{xy}=\frac{\tau_{xy}}{G}=0$$

$$\varepsilon_\theta=100\mu$$

$$\varepsilon_{60^\circ}=\frac{\varepsilon_x+\varepsilon_y}{2}+\frac{\varepsilon_x-\varepsilon_y}{2}\cos2\theta+\frac{\gamma_{xy}}{2}\sin2\theta=100\mu$$

$$\Rightarrow\frac{31.25P+109.375P}{2}+\frac{31.25P-109.375P}{2}\cos2\times(60^\circ)+\frac{\gamma_{xy}}{2}\sin2\times(60^\circ)=100\mu$$

$$\Rightarrow P=1.97(MPa)$$

此時

$$\sigma_y=\frac{PR}{t}=\frac{P\times125}{2\times5}=25P=49.25(MPa)>30(MPa)$$ 塑性變形

(二)材料塑性變形

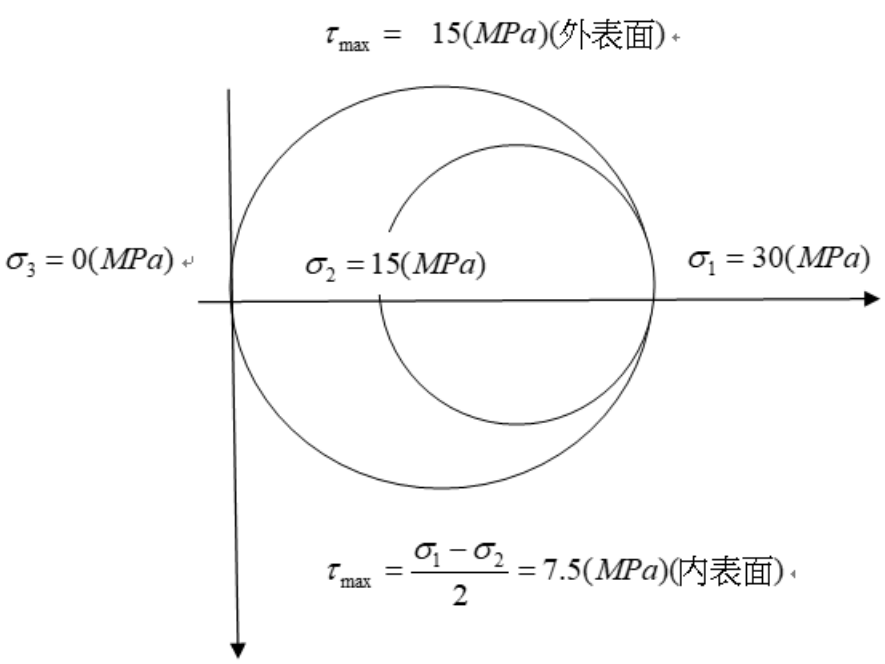
$$\sigma_y = \frac{PR}{t} = \frac{P \times 125}{2 \times 5} = 25P = 30(MPa) \Rightarrow \mathrm{P} = 1.2(MPa)$$

$$\sigma_x = \frac{PR}{2t} = \frac{P \times 125}{2 \times 5} = 12.5P = 15(MPa)$$

$\sigma_1 = 30(MPa)$ 、 $\sigma_2 = 15(MPa)$ 、 $\sigma_3 = 0(MPa)$

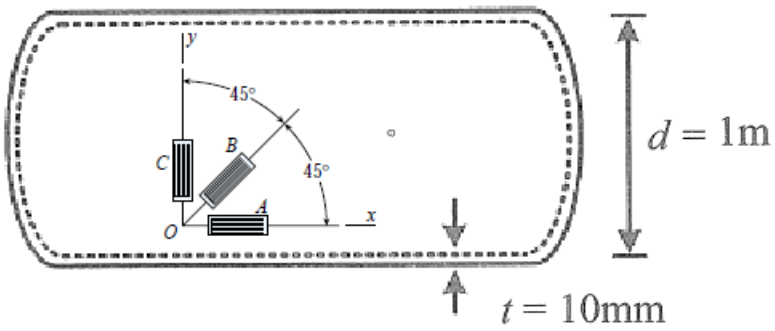
$$\tau_{\max} = \frac{\sigma_1 - 0}{2} = 15(MPa)(\text{外表面})$$

$$\tau_{\max} = \frac{\sigma_1 - \sigma_2}{2} = 7.5(MPa)(\text{內表面})$$



例 19

The pressure in a cylindrical tank as shown in the figure is monitored by a 45° strain rosette mounted on its surface. Unfortunately, during service, gages a and c have become in operative. Only gage b gives a strain reading of 100×10^{-6}. If $d = 1\text{m}$, $t = 10\text{mm}$, $E = 207\text{GPa}$, and $\nu = 0.3$, find the pressure p in the tank.



【台大機械】

【解析】

(一)

$$\sigma_x = \frac{PR}{2t} = \sigma_a$$

$$\sigma_y = \frac{PR}{t} = 2\sigma_a \text{ 、 } \sigma_z = 0$$

$$\varepsilon_x = \frac{1}{E}\times[\sigma_x - \upsilon(\sigma_y + \sigma_z)] = \frac{\sigma_a}{E}\times(1-2\nu)$$

$$\varepsilon_y = \frac{1}{E}\times[\sigma_y - \upsilon(\sigma_x + \sigma_z)] = \frac{\sigma_a}{E}\times(2-\nu)$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G} = 0$$

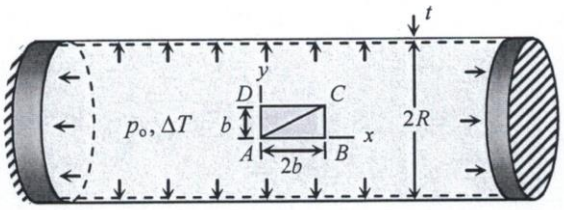
$$\varepsilon_{45^{\circ}} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta = \frac{3(1-\nu)}{2E}\sigma_a$$

帶入數值得 $\sigma_a = 19.71(MPa)$

$$\frac{PR}{2t} = \sigma_a \Rightarrow P = 788.6(KPa)$$

例 20

A rectangular element *ABCD* is drawn on the outer surface of the cylindrical portion of a thin-wall pressure vessel of mean radius *R* and a wall thickness *t* as shown. The initial dimensions of the element are $2b \times b$. This cylindrical pressure vessel is adhered to two rigid walls at its ends and is initially free of stress. Consider the vessel is subjected to an internal pressure p_o and a temperature rise ΔT. As a result of deformation the rectangular element becomes $A^*B^*C^*D^*$. The material constants and the coefficient of thermal expansion are *E*, ν and α. Determine the diagonal length A^*C^* of the deformed element in terms of p_o, ΔT, *E*, ν, α, *R*, *t*, and *b*. (20%)



【104 交大機械】

【解析】

$$\sigma_x = \frac{P_0 R}{2t} \text{ 、 } \sigma_y = \frac{P_0 R}{t} \text{ 、 } \sigma_z = 0$$

$$\varepsilon_x = \frac{1}{E} \times [\sigma_x - \upsilon(\sigma_y + \sigma_z)] + \alpha \Delta T = \frac{P_0 R}{2Et}(1 - 2\upsilon) + \alpha \Delta T$$

$$\varepsilon_y = \frac{1}{E} \times [\sigma_y - \upsilon(\sigma_x + \sigma_z)] + \alpha \Delta T = \frac{P_0 R}{2Et}(2 - \upsilon) + \alpha \Delta T$$

$$\gamma_{xy} = 0$$

$$\varepsilon_{AC} = \frac{\varepsilon_x + \varepsilon_y}{2} + \frac{\varepsilon_x - \varepsilon_y}{2} \cos 2\theta + \frac{\gamma_{xy}}{2} \sin 2\theta$$

將式子帶入

$$\frac{1}{2}[\frac{P_0 R}{2Et}(1 - 2\upsilon) + \alpha \Delta T + \frac{P_0 R}{2Et}(2 - \upsilon) + \alpha \Delta T] + \frac{1}{2}[\frac{P_0 R}{2Et}(1 - 2\upsilon) + \alpha \Delta T - \frac{P_0 R}{2Et}(2 - \upsilon) + \alpha \Delta T]$$

$$\times \cos 2 \times (26.56^\circ) = \frac{3P_0 R}{10Et}(2 - 3\upsilon) + \alpha \Delta T$$

$$\varepsilon_{AC} = \frac{3P_0 R}{10Et}(2 - 3\upsilon) + \alpha \Delta T$$

$$(\overline{AC})' = (\overline{AC}) \times (1 + \varepsilon_{AC}) = \sqrt{5} b \times [1 + \frac{3P_0 R}{10Et}(2 - 3\upsilon) + \alpha \Delta T]$$